Probability of Default

The **Probability of Default (PD)** is a key metric in credit risk management that estimates the likelihood that a borrower will fail to meet their debt obligations over a specified time period, typically one year. It is expressed as a percentage, where a higher PD indicates a greater risk of default.

Key Aspects of Probability of Default (PD):

1. **Use in Risk Assessment**:
    - Financial institutions use PD to quantify credit risk and assess the overall quality of their credit portfolio.
    - It helps lenders determine terms of credit, such as interest rates and credit limits, based on a borrower's risk level.
2. **Calculation**:
    - PD can be calculated using various models that analyze historical data, credit scores, macroeconomic conditions, and borrower-specific financial metrics (e.g., income, employment status, financial history).
    - **Examples**: Logistic regression, machine learning models, and Z-score can all be used to predict PD.
3. **Factors Influencing PD**:
    - **Credit History**: Poor credit history (e.g., missed payments, high credit utilization) increases PD.
    - **Economic Environment**: During economic downturns, PD increases as unemployment rises and financial stability declines.
    - **Borrower-Specific Variables**: Income stability, debt-to-income ratio, age, and even sector-specific risk for business loans are considered.
4. **Application in Credit Decisions**:
    - A high PD may lead to loan rejection, higher interest rates, or stricter terms.
    - For lenders, accurately estimating PD is essential for capital adequacy requirements, as it directly affects risk-weighted assets and regulatory capital.
5. **Role in Loan Loss Provisioning**:
    - PD is crucial for estimating **Expected Credit Loss (ECL)**, a measure that combines PD with other factors like Loss Given Default (LGD) and Exposure at Default (EAD) to calculate the potential losses a lender may face.

Examples of PD Usage in Models:

- **FICO Score**: A borrower with a low FICO score has a high PD, implying they're at greater risk of default.
- **KMV Model**: For corporate borrowers, PD is determined by assessing the firm's assets relative to liabilities.
- **Credit Scoring Models**: Use historical data on defaults across borrower profiles to set cut-off scores that reflect acceptable PD levels.